CATFISH tutorial

Pathway analysis and candidate genes from GWAS summary statistics

Contents

1. What CATFISH does **1**

2. Installation **2**

3. Quick start on simulated data **2**

4. Everything in one call: `catfish_run()` **3**
4.1 Starting from GWAS summary statistics . . . 3
4.2 Starting from MAGMA output . . . 3
4.3 What it does, and what it returns . . . 3

5. Step by step **4**
5.1 Gene locations . . . 4
5.2 MAGMA annotation (SNPs to genes) . . . 4
5.3 MAGMA gene analysis . . . 4
5.4 Gene results and gene–gene correlations . . . 5
5.5 Pathways . . . 5
5.6 The five component tests, one at a time . . . 5
5.7 Westfall–Young pathway test . . . 6
5.8 Components and calibration together: `catfish_omni2_pathways()` . . . 7
5.9 Candidate genes . . . 7

6. Choosing the settings **7**

7. Output columns **8**
Westfall–Young (`catfish_wy_pathways()`, `res$pathways`) . . . 8
`catfish_omni2_pathways()` . . . 8
`gene_prioritize()` . . . 9

8. Function reference **9**

1. What CATFISH does

CATFISH takes GWAS summary statistics, turns them into gene-level results with MAGMA, and tests pathways. Every pathway is scored by five tests that suit different signal shapes (ACAT, Fisher, adaptive soft TFisher, minP, Stouffer), combined into one omnibus with ACAT. The omnibus is calibrated by a Westfall–Young procedure: each null replicate draws one genome-wide vector of correlated gene z-scores, chromosome by chromosome, from the MAGMA gene–gene correlations, and scores every pathway on that same draw. Each pathway gets a calibrated p-value and a family-wise p-value that controls the error rate over all pathways tested. Finally, genes are ranked as candidates from SNP, gene and pathway evidence.

There are two ways to run it:

	Function	When
Everything in one call	`catfish_run()`	From GWAS summary statistics, or from MAGMA output you already have, to pathway results and candidate genes
Step by step	`magma_annotate()`, `magma_gene()`, `magma_read_genes()`, `magma_genesraw_to_cor_pairs_banded()`, `catfish_load_pathways()`, the five component tests, `catfish_wy_pathways()`, `catfish_omni2_pathways()`, `gene_prioritize()`	To run one part, change a setting of one step, or split MAGMA into outer jobs

`catfish_run()` calls exactly the step functions, so both ways give the same results with the same settings.

The GWAS itself (GEMMA, GAPIT, PLINK, …) is run outside CATFISH.

2. Installation

```
install.packages("remotes")
remotes::install_github("nirwan1265/CATFISH")
```

This installs the required packages (ACAT from GitHub, TFisher and data.table from CRAN). Optional packages:

```
install.packages(c("metap", "evir", "BiocManager"))
BiocManager::install(c("qvalue", "IHW", "rtracklayer", "GenomicRanges", "S4Vectors"))
```

MAGMA is a separate program (https://ctg.cncr.nl/software/magma). It is needed only to run MAGMA from R:

```
library(CATFISH)
options(magma.path = "/full/path/to/magma")
magma_path()      # checks that it is found
```

3. Quick start on simulated data

This runs in a few seconds and needs no MAGMA. It simulates 300 genes on three chromosomes with neighbouring genes correlated, and puts six associated genes into one pathway.

```
library(CATFISH)
set.seed(1)
genes <- data.frame(GENE = paste0("g", 1:300), CHR = rep(1:3, each = 100))
R <- diag(100)
R[cbind(1:99, 2:100)] <- R[cbind(2:100, 1:99)] <- 0.3
z <- as.vector(replicate(3, t(chol(R)) %*% rnorm(100)))
z[1:6] <- z[1:6] + 4
genes$ZSTAT <- z
genes$P     <- pnorm(z, lower.tail = FALSE)
cor_pairs <- do.call(rbind, lapply(split(genes$GENE, genes$CHR), function(g)
  data.frame(gene1 = head(g, -1), gene2 = tail(g, -1), r = 0.3)))
pathways <- c(list(PWY_signal = paste0("g", 1:15)),
              setNames(lapply(1:19, function(i) sample(genes$GENE[-(1:15)], 15)),
                       paste0("PWY_", 1:19)))

res <- catfish_wy_pathways(genes, pathways = pathways, cor_pairs = cor_pairs,
```

```
                          B = 200, min_genes = 10, seed = 1)
head(res[, c("pathway_id", "n_genes", "z_std", "p_calibrated", "p_fwer_std")])

cand <- gene_prioritize(magma = genes, pathway_results = res, pathways = pathways)
head(cand[, c("rank", "gene", "score", "magma_rank", "pathway_rank", "best_pathway")])
```

Only `PWY_signal` should have `p_fwer_std <= 0.05`, and the six associated genes should rank first.

4. Everything in one call: `catfish_run()`

4.1 Starting from GWAS summary statistics

```
res <- catfish_run(
  gwas_file      = "dry_tons.assoc.txt",          # GEMMA output, one SNP per row
  rename_columns = c(CHR = "chr", SNP = "rs", POS = "ps",
                     PVALUE = "p_wald", NMISS = "n_miss"),
  n_total        = 327,                           # lines in the GWAS
  bfile          = "bap_genotypes",               # PLINK LD reference
  species        = "sorghum",                     # built-in gene locations and pathways
  window         = c(25, 25),                     # kb around each gene
  gene_model     = "snp-wise=mean",
  chroms         = 1:10, n_threads = 4,           # one MAGMA run per chromosome
  B              = 10000,
  min_genes      = 5, max_genes = 200,
  out_dir        = "dry_tons", out_prefix = "dry_tons"
)
```

`rename_columns` maps CATFISH's names to your file's columns: `CHR`, `SNP`, `POS`, `PVALUE` are required; the sample size is either a per-SNP column (`N`, `NOBS`, or `NMISS` together with `n_total`) or `n_total` alone.

4.2 Starting from MAGMA output

On a cluster, MAGMA is usually run one chromosome per job (section 5.3). Point `catfish_run()` at the folder of `.genes.out` and `.genes.raw` files; MAGMA is then not run:

```
res <- catfish_run(
  magma_dir     = "magma",
  magma_pattern = "^dry_tons_chr.*\\.genes\\.out$",
  species       = "sorghum",
  gwas_file     = "dry_tons.assoc.txt",          # optional: SNP layer of the gene ranking
  rename_columns = c(CHR = "chr", SNP = "rs", POS = "ps", PVALUE = "p_wald"),
  gene_annot    = "magma/dry_tons.genes.annot",  # optional: SNP-to-gene map for that layer
  out_dir       = "dry_tons"
)
```

4.3 What it does, and what it returns

Step	Done by	Notes
1	`magma_annotate()`, `magma_gene()`	only with `gwas_file` and no `magma_dir`
2	`magma_read_genes()`	combines all chromosomes; prints the median of every gene p column
3	`magma_genesraw_to_cor_pairs_banded()`	gene–gene correlations of all chromosomes in one file; reused on a rerun

Step	Done by	Notes
4	`catfish_wy_pathways()`	Westfall–Young pathway test
5	`catfish_omni2_pathways()`, analytic only	the five component p-values, added to the pathway table (`components = TRUE`)
6	`gene_prioritize()`	candidate genes (`prioritize = TRUE`)

The gene p-value column is `P_SNPWISE_MEAN` when present (a `multi=snp-wise` run) and `P` otherwise; set `p_col` to choose another.

`res` is a list:

Element	Content
`res$pathways`	one row per pathway: Westfall–Young results plus `acat_p`, `fisher_p`, `tfisher_p`, `minp_p`, `stouffer_p`
`res$candidates`	ranked genes
`res$genes`	gene-level results used
`res$pathway_sets`	pathway memberships used
`res$settings`	`p_col`, `gene_regex`, `B`, `tau_grid`, …
`res$files`	everything written to `out_dir` (`<out_prefix>_pathways.csv`, `<out_prefix>_candidate_genes.csv`, the correlation file, MAGMA output)

5. Step by step

5.1 Gene locations

```
gff3_to_geneloc(gff = "reference.gff3", out = "species.genes.loc")
```

Built-in files exist for maize, sorghum and arabidopsis (`species =` in `magma_annotate()` and `catfish_run()`). Non-numeric chromosome names can be recoded to numbers (`recode_chr = "auto"`), with a map file written beside the output.

5.2 MAGMA annotation (SNPs to genes)

```
ann <- magma_annotate(
  stats_file     = "dry_tons.assoc.txt",
  rename_columns = c(CHR = "chr", SNP = "rs", POS = "ps", PVALUE = "p_wald"),
  gene_loc       = "species.genes.loc",
  out_prefix     = "dry_tons",
  out_dir        = "magma",
  window         = c(25, 25),
  nonhuman       = TRUE
)
ann$gene_annot     # the .genes.annot file for the next step
```

5.3 MAGMA gene analysis

```
magma_gene(
  bfile          = "bap_genotypes",
  gene_annot     = ann$gene_annot,
```

```
  stats_file     = "dry_tons.assoc.txt",
  rename_columns = c(CHR = "chr", SNP = "rs", PVALUE = "p_wald", NMISS = "n_miss"),
  n_total        = 327,
  out_prefix     = "dry_tons",
  out_dir        = "magma",
  gene_model     = "snp-wise=mean",
  chroms         = 1:10,       # writes dry_tons_chr1.snp_wise_mean.genes.out, ...
  n_threads      = 4           # 1 = one chromosome after another
)
```

For a cluster array job, give the job's chromosome: `chroms = 5`.

5.4 Gene results and gene–gene correlations

```
genes <- magma_read_genes(dir = "magma", pattern = "^dry_tons_chr.*\\.genes\\.out$")

raw <- sub("\\.genes\\.out$", ".genes.raw", attr(genes, "files"))
magma_genesraw_to_cor_pairs_banded(raw, "dry_tons_gene_cor_pairs.txt",
                                   gene_regex = "^SORBI")
```

`magma_read_genes()` prints the median of every gene p-value column; about 0.5 is expected where most genes carry no signal (see section 6). MAGMA computes gene–gene correlations only within a chromosome, so all chromosome files are needed. `gene_regex` must match your gene IDs: `"^SORBI"`, `"^Zm"`, `"^AT"`, `"^FBgn"`, `"^[0-9]+$"` (Entrez).

5.5 Pathways

```
pw <- catfish_load_pathways(species = "sorghum")
# built in: "maize", "sorghum", "arabidopsis", "plant", "fly", "human", "human_reactome"

pw <- catfish_load_pathways(file = "sorghumbicolorcyc_pathways.tsv")   # your PMN file
pw <- catfish_load_gmt("h.all.v2024.1.Hs.entrez.gmt")                  # GMT
pw <- catfish_load_biocyc("humancyc_export.csv", gene_col = "NCBI Gene ID")
```

A pathway set is either a data.frame with `pathway_id`, `gene_id` (and optionally `pathway_name`) or a named list of gene vectors. Gene IDs are matched to the gene results without regard to case. `catfish_write_set_annot()` writes a set as a MAGMA `--set-annot` file.

5.6 The five component tests, one at a time

Each takes the gene results and the pathways, and returns one row per pathway. These p-values are analytic: they treat genes as independent, so with correlated genes they are anti-conservative. Use them to see which kind of signal drives a pathway; use the Westfall–Young results (5.7) for inference.

Test	Function	Sensitive to	Statistic
ACAT	`catfish_acat_pathways()`	a few very strong genes	Cauchy combination of gene p-values
Fisher	`catfish_fisher_pathways()`	many moderate genes	-2 sum log p
Adaptive soft TFisher	`catfish_soft_tfisher_adaptive_pathways()`	strong genes or many moderate support	Fisher on p-values below a threshold tau, best tau of a grid
minP	`catfish_minp_pathways()`	one dominant gene	smallest gene p, Sidak-corrected
Stouffer	`catfish_stoufferZ_pathways()`	weak shift across the whole pathway	sum of gene z-scores / sqrt(n)

```
acat  <- catfish_acat_pathways(genes, pathways = pw, gene_col = "GENE", p_col = "P",
                               min_p = 1e-15)
fish  <- catfish_fisher_pathways(genes, pathways = pw, p_col = "P")
tfish <- catfish_soft_tfisher_adaptive_pathways(genes, pathways = pw, p_col = "P",
                                                tau_grid = c(0.01, 0.05, 0.5))
minp  <- catfish_minp_pathways(genes, pathways = pw, p_col = "P")
stou  <- catfish_stoufferZ_pathways(genes, pathways = pw, z_col = "ZSTAT",
                                    weight_col = NULL, alternative = "greater")
```

Parameters they share:

Parameter	Meaning
`gene_results`	gene table (`GENE`, `P`, …)
`pathways` / `species`	pathway set, or a built-in one
`pmn_gene_col`	gene column of a built-in file
`gene_col`, `p_col`	gene ID and p-value columns (Stouffer: `z_col`)
`min_p`	p-values below this are set to it before combining (default `1e-15`)
`do_fix`	clean p-values to lie strictly in (0, 1) (default `TRUE`)
`output`, `out_dir`	write a CSV

Test-specific parameters:

Function	Parameter	Meaning
soft TFisher	`tau_grid`	thresholds; a numeric vector, or `"auto"` (quantiles of all gene p-values)
Stouffer	`z_col`	gene z column (default `ZSTAT`)
Stouffer	`weight_col`	optional per-gene weights (e.g. sqrt(NSNPS)); default equal
Stouffer	`min_abs_w`	replaces missing or non-positive weights
Stouffer	`alternative`	`"greater"` (enrichment; default), `"two.sided"`, `"less"`

Output columns: `pathway_id`, `pathway_name`, `n_genes`, `gene_names`, and the test's own: `acat_p`; `fisher_p`; `tau_hat`, `tfisher_stat_hat`, `tfisher_p_hat`, `tfisher_p_min`, `tfisher_p_omni`; `minp_stat`, `minp_p`, `minp_method`; `stouffer_Z`, `stouffer_p`.

5.7 Westfall–Young pathway test

```
res <- catfish_wy_pathways(
  gene_results = genes,
  pathways     = pw,
  gene_col     = "GENE", p_col = "P", chr_col = "CHR",
  cor_file     = "dry_tons_gene_cor_pairs.txt",
  B            = 10000,
  use_tfisher  = TRUE,
  tau_grid     = c(0.01, 0.05, 0.5),
  min_genes    = 5, max_genes = 200,
  seed         = 20260706
)
attr(res, "wy_threshold_05")     # omnibus p needed for FWER 5% on the raw scale
```

The omnibus combines ACAT, Fisher, Stouffer, minP and (with `use_tfisher`) soft TFisher with ACAT, all in log space, so gene p-values far below 1e-300 keep their order. `tau_grid` must be fixed in advance: a grid estimated from the gene p-values would make each pathway's statistic depend on genes outside it and break the family-wise control. Without `cor_file` (or `cor_pairs`) every gene is treated as independent, and CATFISH warns.

5.8 Components and calibration together: `catfish_omni2_pathways()`

The same Westfall–Young calibration, with the component p-values, multiple testing adjustments and diagnostics in one table:

```
om <- catfish_omni2_pathways(
  gene_results = genes, pathways = pw,
  gene_col = "GENE", p_raw_col = "P", z_col = "ZSTAT", chr_col = "CHR",
  min_genes = 5, max_genes = 200,
  perm_mode = "wy", B_perm = 10000,
  wy_tau_grid = c(0.01, 0.05, 0.5),
  magma_cor_file = "dry_tons_gene_cor_pairs.txt",
  seed = 20260706
)
```

`perm_mode = "none"` gives the analytic table only; `"global"` resamples genes from a pooled distribution (a faster heuristic that ignores gene correlation); `"both"` runs both. With `perm_mode = "wy"`, `omni_p_cal` and `omni_p_fwer_wy_std` equal `p_calibrated` and `p_fwer_std` of `catfish_wy_pathways()` for the same seed.

5.9 Candidate genes

```
cand <- gene_prioritize(
  magma           = genes,  magma_p_col = "P",
  pathway_results = res,                        # from 5.7
  pathways        = pw,
  gwas            = "dry_tons.assoc.txt",       # optional SNP layer
  gwas_snp_col    = "rs", gwas_p_col = "p_wald",
  gene_annot      = ann$gene_annot,             # SNP-to-gene map (or gene_loc + window)
  targets         = c("GENE_A", "GENE_B")       # optional known genes
)
attr(cand, "targets")     # where the known genes rank
```

Each gene's score is

```
score = w_gwas * S_gwas + w_magma * S_magma + w_pathway * S_pathway
        + w_hub * log10(1 + n_pathways)
```

with default weights 1, 1, 0.5 and 0.3. Each S is a rank score, -log10(rank / N): the strongest SNP in or near the gene, the MAGMA gene p-value, and the strongest tested pathway containing the gene (by `z_std`). SNPs are assigned to genes by `gene_annot` (the MAGMA annotation) or, failing that, by position within `window` bp of each gene in `gene_loc`. Give the SNP layer as -log10(p) (`gwas_neglog10p_col`) when the p column is floored (e.g. at 1e-300), since a floor ties the strongest genes.

6. Choosing the settings

Setting	Recommendation	Why
MAGMA gene model	`snp-wise=mean`	The Westfall–Young null assumes gene p-values are uniform where there is no association. The SNP-wise mean model is calibrated when the GWAS is; the SNP-wise top model, and `multi=snp-wise` which includes it, are often inflated in panels with long-range LD (median gene p of 0.11 and 0.18 against 0.45 in a sorghum panel), and inflation is read as pathway signal. Check the medians printed by `magma_read_genes()`.
`window`	25 kb in plants with extended LD; smaller (a few kb) where LD decays fast	SNPs outside genes are assigned to genes within the window.
`B`	10,000 for final results; 200–1,000 to try settings	The smallest calibrated p is about 1/(B+1).
`tau_grid`	`c(0.01, 0.05, 0.5)`	Fixed in advance, as required for family-wise control.
`min_genes`, `max_genes`	5–10 and 200	Very small pathways are dominated by one gene; very large ones are unspecific.
`seed`	any fixed integer	Makes the null draws reproducible.

Gene-length adjustment (`catfish_adjust_gene_p()`, `get_gene_lengths()`) is not needed for MAGMA gene p-values, which are calibrated for gene size. It is for gene p-values from methods that are not, such as an uncorrected minimum SNP p per gene.

7. Output columns

Westfall–Young (`catfish_wy_pathways()`, `res$pathways`)

Column	Meaning
`pathway_id`, `pathway_name`	pathway
`z_std`	observed omnibus standardised against the pathway's own null; more negative is stronger. Results are sorted by it.
`p_fwer_std`	Westfall–Young family-wise p-value; the headline result
`p_calibrated`	per-pathway calibrated p-value
`n_genes`	genes of the pathway present in the gene results
`log10_p_omnibus`, `p_omnibus`	analytic (uncalibrated) omnibus
`p_fwer_wy`	family-wise p-value on the raw omnibus scale
`p_bonferroni`	Bonferroni on the analytic omnibus, for comparison
`acat_p`, `fisher_p`, `tfisher_p`, `minp_p`, `stouffer_p`	analytic component p-values (`catfish_run()` only)

`catfish_omni2_pathways()`

Column	Meaning
`acat_p`, `fisher_p`, `tfisher_p_analytic`, `minp_p_analytic`, `stouffer_p_analytic`	analytic components
`omni_p_analytic`	analytic omnibus
`omni_p_cal`, `omni_p_fwer_wy_std`, `omni_p_fwer_wy`, `omni_z_std`	Westfall–Young results (as above)
`omni_p_global`	global-resampling p-value (`perm_mode = "global"` or `"both"`)
`omni_p_final`	best available: Westfall–Young calibrated, else global, else analytic
`*_BH`, `omni_p_final_q_storey`, `omni_p_final_q_bky10`, `omni_p_final_q_ihw10`	multiple-testing adjustments
`dominant_component`	component with the smallest p-value
`agreement_score`, `agreement_k`, `agreement_m`	fraction / number of components with $p < 0.05$
`genes_used`, `gene_pvals_used`	genes of the pathway and their p-values

`gene_prioritize()`

Column	Meaning
`rank`, `gene`, `symbol`, `score`	final rank and score
`n_layers`	layers (SNP, gene, pathway) that support the gene
`gwas_score`, `magma_score`, `pathway_score`, `hub_bonus`	the parts of the score
`gwas_neglog10p`, `gwas_rank`	strongest SNP of the gene
`magma_p`, `magma_rank`	MAGMA gene p-value and its rank
`best_pathway`, `pathway_stat`, `pathway_rank`	strongest tested pathway containing the gene
`n_pathways`	tested pathways containing the gene

8. Function reference

Every exported function with all of its parameters, taken from the package documentation (`?function_name` in R shows the same).

`catfish_run()`

Run the whole CATFISH analysis in one call

```
catfish_run(
  gwas_file = NULL,
  rename_columns = NULL,
  bfile = NULL,
  gene_loc = NULL,
  n_total = NULL,
  window = c(25, 25),
  gene_model = "snp-wise=mean",
  chroms = NULL,
  n_threads = 1L,
  nonhuman = TRUE,
  sep = NULL,
  chr_map_path = NULL,
  magma_dir = NULL,
  magma_pattern = NULL,
  gene_annot = NULL,
  pathways = NULL,
  species = NULL,
```

```
  pmn_gene_col = NULL,
  p_col = NULL,
  gene_regex = NULL,
  B = 10000L,
  use_tfisher = TRUE,
  tau_grid = c(0.01, 0.05, 0.5),
  min_genes = 10L,
  max_genes = 200L,
  seed = 20260706L,
  components = TRUE,
  prioritize = TRUE,
  targets = NULL,
  w_gwas = 1,
  w_magma = 1,
  w_pathway = 0.5,
  w_hub = 0.3,
  out_dir = "catfish_results",
  out_prefix = "catfish",
  verbose = TRUE
)
```

Parameter	Description
`gwas_file`	GWAS summary statistics file (one SNP per row).
`rename_columns`	Named character vector mapping the standard names to the columns of `gwas_file`: `CHR`, `SNP`, `POS`, `PVALUE`, and optionally a per-SNP sample size (`N`, `NOBS`, or `NMISS` with `n_total`). Example: `c(CHR = "chr", SNP = "rs", POS = "ps", PVALUE = "p_wald", NMISS = "n_miss")`.
`bfile`	PLINK prefix of the LD reference genotypes (`.bed/.bim/.fam`).
`gene_loc`	MAGMA gene-location file. If `NULL`, the built-in file of `species` is used (maize, sorghum, arabidopsis).
`n_total`	Total sample size, when `rename_columns` has no per-SNP `N`/`NOBS`, or with `NMISS`.
`window`	SNP-to-gene window in kb, `c(upstream, downstream)`.
`gene_model`	MAGMA gene model (one). `"snp-wise=mean"` is recommended: see Details.
`chroms`	Chromosomes to analyse one MAGMA run at a time (e.g. `1:10`). `NULL` runs MAGMA once on the whole genome.
`n_threads`	Parallel MAGMA runs when `chroms` is given.
`nonhuman`	Passed to MAGMA annotation (`TRUE` for non-human data).
`sep`	Field separator of `gwas_file`; inferred from the extension if `NULL`.
`chr_map_path`	Optional chromosome-renaming map (see `gff3_to_geneloc`).
`magma_dir`	Folder with existing MAGMA output (`.genes.out` and `.genes.raw`).
`magma_pattern`	Regular expression selecting the `.genes.out` files in `magma_dir`; default all of them.

Parameter	Description
gene_annot	Optional MAGMA .genes.annot for the SNP layer of the candidate gene ranking when starting from magma_dir with a gwas_file. Without it, SNPs are assigned to genes by position.
pathways	Pathways: named list of gene vectors, or data.frame with pathway_id, gene_id (optional pathway_name). Give this or species.
species	Built-in pathways (and gene locations), e.g. "sorghum"; see catfish_load_pathways.
pmn_gene_col	Gene column of a built-in pathway file.
p_col	Gene p-value column. Default: P_SNPWISE_MEAN when present (a multi=snp-wise run), otherwise P.
gene_regex	Regular expression identifying gene IDs in the .genes.raw files; worked out from the gene IDs if NULL.
B	Westfall–Young null replicates.
use_tfisher	Include soft TFisher in the omnibus.
tau_grid	Fixed soft TFisher grid (also used for the component table).
min_genes, max_genes	Pathway size limits (genes present).
seed	Seed for the null draws.
components	Add the five analytic component p-values to the pathway table.
prioritize	Rank candidate genes.
targets	Optional known genes; their ranks are reported.
w_gwas, w_magma, w_pathway, w_hub	Layer weights of gene_prioritize.
out_dir	Output folder.
out_prefix	Prefix of the output files.
verbose	Print progress.

gff3_to_geneloc()

Convert a GFF3 file to a MAGMA gene location file (optionally recoding CHR to 1..K)

```
gff3_to_geneloc(
  gff,
  out,
  feature_type = "gene",
  id_fields = c("gene_id", "ID", "Name"),
  drop_scaffolds = TRUE,
  scaffold_regex = "^(scaf|scaffold|un|unplaced|chrUn|KI|GL|NW_|NT_)",
  chr_prefix = "chr",
  recode_chr = c("auto", "none", "order", "map"),
  chr_order = NULL,
  chr_map = NULL,
  strict_chr = FALSE,
  write_chr_map = TRUE,
  chr_map_out = NULL
)
```

Parameter	Description
gff	Path to a GFF3 file.
out	Path to write the MAGMA gene location file (tab-delimited).
feature_type	GFF3 feature type(s) to use (default “gene”).

Parameter	Description
id_fields	Candidate attribute tags to use as gene IDs (searched in order), e.g. c("gene_id","ID","Name").
drop_scaffolds	Logical; if TRUE, drop seqids that look like scaffolds.
scaffold_regex	Regex for scaffold/unplaced contigs to drop when drop_scaffolds=TRUE.
chr_prefix	Optional string to strip from start of seqid when reading (e.g. "chr").
recode_chr	One of "none","auto","order","map".
chr_order	If recode_chr="order", the desired order of original chromosome labels.
chr_map	If recode_chr="map", a named vector mapping original->new numeric labels.
strict_chr	If TRUE, error when encountering CHR labels not in chr_order/chr_map.
write_chr_map	If TRUE, write a mapping file.
chr_map_out	Optional explicit path for the mapping file. Default: paste0(out, ".chr_map.tsv")

magma_path()

Locate MAGMA binary

```
magma_path()
```

magma_annotate()

Run MAGMA SNP-to-gene annotation

```
magma_annotate(
  stats_file,
  rename_columns,
  gene_loc = NULL,
  out_prefix,
  out_dir = NULL,
  window = NULL,
  species = NULL,
  sep = NULL,
  nonhuman = NULL,
  filter_snps = NULL,
  chr_map_path = NULL,
  strict_chr = TRUE,
  verbose = TRUE
)
```

Parameter	Description
stats_file	Path to the GWAS summary statistics file.
rename_columns	Named character vector mapping required MAGMA columns to the column names in `stats_file`; must contain at least `CHR`, `SNP`, and `POS` (e.g. `c(CHR = "chr", SNP = "rs", POS = "ps", PVALUE = "p")`).
gene_loc	Path to a MAGMA gene-location file (e.g. from `gff3_to_geneloc`).
out_prefix	Output file prefix.
out_dir	Optional output directory. Created if it does not exist.

Parameter	Description
`window`	Optional numeric vector `c(up, down)` giving the upstream/downstream window in kb around each gene.
`species`	Optional species tag (used to pick a built-in gene-location file when `gene_loc` is not supplied).
`sep`	Field separator for `stats_file`. If `NULL`, inferred from the file extension.
`nonhuman`	Logical or `NULL`; set `TRUE` for non-human data so MAGMA does not assume human chromosome coding.
`filter_snps`	Optional vector or file of SNPs to restrict the analysis to.
`chr_map_path`	Optional path to a two-column chromosome-renaming map.
`strict_chr`	Logical; if `TRUE`, error when the chromosome map is missing an entry rather than passing the value through unchanged.
`verbose`	Logical; print progress messages.

`magma_gene()`

Run MAGMA gene analysis on SNP p-values

```
magma_gene(
  bfile,
  gene_annot,
  pval_file = NULL,
  stats_file = NULL,
  n_total,
  snp_col = "SNP",
  p_col = "P",
  rename_columns = NULL,
  out_prefix,
  out_dir = NULL,
  gene_model = "snp-wise=top",
  chroms = NULL,
  n_threads = 1,
  chr_keep = NULL,
  sep = NULL,
  chr_map_path = NULL,
  strict_chr = TRUE,
  verbose = TRUE
)
```

Parameter	Description
`bfile`	PLINK prefix for LD reference.
`gene_annot`	MAGMA gene annotation file (.genes.annot).
`pval_file`	Optional; SNP p-value file for MAGMA's –pval. If NULL, `stats_file` must be provided.
`stats_file`	Optional; GWAS summary statistics file containing at least SNP ID and p-value columns. If provided, `pval_file` is ignored and a MAGMA p-value file is generated internally.

Parameter	Description
n_total	Total sample size (N=… in MAGMA) when using a constant N or computing per-SNP N from NMISS. Not required if rename_columns supplies an N or NOBS column.
snp_col	Column name containing SNP IDs (used only when pval_file is given and rename_columns is NULL).
p_col	Column name containing p-values (used only when pval_file is given and rename_columns is NULL).
rename_columns	Optional named character vector mapping standard names to the column names in stats_file, e.g. c(CHR = "chr", SNP = "rs", POS = "ps", PVALUE = "p_wald", NMISS = "n_miss") or c(..., N = "N_eff") or c(..., NOBS = "Nobs"). For magma_gene(), only CHR, SNP, PVALUE, N, NOBS, and NMISS are used.
out_prefix	Base output prefix (file name without directory).
out_dir	Optional output directory. If not NULL, the final prefix will be file.path(out_dir, out_prefix). The directory is created if it does not exist. For stats_file mode, the internal p-value file is also written into this directory.
gene_model	Character vector of one or more MAGMA gene models, e.g. “snp-wise=top”, “snp-wise=mean”, “multi”.
chroms	Optional vector of chromosomes to analyse separately (e.g. 1:10), each written as <out_prefix>_chr<k>. Only used in stats_file mode. With n_threads > 1 the chromosomes run in parallel, otherwise one after another. In a cluster array job give the job’s single chromosome, e.g. chroms = 5.
n_threads	Number of parallel workers to use when chroms is provided. Defaults to 1 (no parallelism). Will be clamped to min(length(chroms), detectCores() - 1).
chr_keep	Internal. Optional subset of chromosomes to retain.
sep	Field separator for stats_file: ",", " ", or a tab. If NULL, inferred from the file extension.
chr_map_path	Optional path to a two-column chromosome-renaming map.
strict_chr	Logical; if TRUE, error when the chromosome map is missing an entry rather than passing the value through unchanged.
verbose	Logical; print progress messages.

magma_read_genes()

Read and combine MAGMA gene results

```
magma_read_genes(
  files = NULL,
  dir = NULL,
  pattern = "\\.genes\\.out$",
  verbose = TRUE
)
```

Parameter	Description
files	Character vector of .genes.out paths. Provide this or dir.
dir	Directory to search for files matching pattern.
pattern	Regular expression for the file names when dir is used.
verbose	Print the number of genes and chromosomes and the median of each gene p-value column.

magma_genesraw_to_cor_pairs_banded()

Extract gene-gene correlations from MAGMA .genes.raw files

```
magma_genesraw_to_cor_pairs_banded(
  genes_raw_file,
  out_pairs_file,
  fixed_fields = NULL,
  gene_regex = "^FBgn",
  keep_abs_r_ge = 0,
  overwrite = TRUE,
  verbose = TRUE
)
```

Parameter	Description
genes_raw_file	Path(s) to MAGMA .genes.raw files. Give one file per chromosome when MAGMA was run chromosome by chromosome; all are written to one output file. MAGMA computes gene–gene correlations only within a chromosome, so the files together hold the full correlation structure.
out_pairs_file	Output path for the correlation pairs file.
fixed_fields	Integer, or NULL (default) to read it from each file's # COVAR header line: the seven fixed MAGMA fields (GENE, CHR, START, STOP, NSNPS, NPARAM, N) plus one per covariate. For MAGMA 1.10 output (# COVAR = NSAMP MAC) this is 9 for every gene model. Used when a file has no # COVAR line.
gene_regex	Regular expression to identify gene IDs. Use organism-specific patterns: • Maize: "^Zm" • Fly: "^FBgn" • Arabidopsis: "^AT" • Sorghum: "^SORBI"
keep_abs_r_ge	Numeric. Only keep correlations with |r| >= this value. Default 0 keeps all correlations.
overwrite	Logical. Overwrite output file if it exists.
verbose	Logical. Print progress messages.

catfish_load_pathways()

Load pathways as a data.frame

```
catfish_load_pathways(
  species = c("maize", "sorghum", "arabidopsis", "plant", "fly", "human",
    "human_reactome"),
  gene_col = NULL,
  drop_unknown = TRUE,
  file = NULL
)
```

Parameter	Description
`species`	Character. One of “maize”, “sorghum”, “arabidopsis”, “plant”, “fly”, “human” (HumanCyc) or “human_reactome” (Reactome, Entrez IDs, from MSigDB).
`gene_col`	Character or NULL. Column name to use for gene IDs. If NULL (default), the function will: • For PMN files: use “Gene-name” if present, else “Gene-id” • For fly: use “gene”
`drop_unknown`	Logical. If TRUE (default), drop rows where gene_id is NA, empty, or “unknown”.
`file`	Optional path to your own pathway file in PMN / Pathway Tools tab-separated format (columns `Pathway-id`, `Pathway-name` and the gene column), e.g. a newer PMN release or another species from https://plantcyc.org. When given, `species` is ignored.

`catfish_pmn_file()`

Path to built-in pathway file

```
catfish_pmn_file(
  species = c("maize", "sorghum", "arabidopsis", "plant", "fly", "human",
    "human_reactome")
)
```

Parameter	Description
`species`	Which built-in database to use: • “maize”: CornCyc pathways (corncyc_pathways.20230103) • “sorghum”: SorghumBicolorCyc pathways with SORBI_ gene IDs, the IDs of the built-in sorghum.genes.loc (sorghumbicolorcyc_pathways.20230103.SORBI) • “arabidopsis”: AraCyc pathways (aracyc_pathways.20230103) • “plant”: PlantCyc general plant pathways (plantcyc_pathways.20230103) • “fly”: FlyCyc Drosophila pathways (Fly_Cyc.tsv) • “human”: HumanCyc pathways (humancyc_pathways.20261024) • “human_reactome”: Reactome pathways for human, Entrez gene IDs (human_reactome.pw; 1,867 pathways from MSigDB v2026.2 C2:CP:REACTOME, c2.cp.reactome.v2026.2.Hs.entrez). MSigDB gene sets are distributed under the Creative Commons Attribution 4.0 International License; cite MSigDB and Reactome when you use them. One pathway per line: ID, then member genes.

`catfish_load_gmt()`

Load pathways from a GMT file

```
catfish_load_gmt(file, min_size = 1L, max_size = Inf, drop_unknown = TRUE)
```

Parameter	Description
`file`	Path to a .gmt file.
`min_size`, `max_size`	Integer. Keep only pathways whose gene count falls within these bounds after de-duplication. Defaults keep everything; 10 and 200 are common choices for GWAS pathway analysis.
`drop_unknown`	Logical. Drop blank and "unknown" gene identifiers.

`catfish_load_biocyc()`

Load pathways from a BioCyc / Pathway Tools SmartTable export

```
catfish_load_biocyc(
  file,
  id_col = "Object ID",
  name_col = "Pathways",
  gene_col = "Genes of Pathways",
  min_size = 1L,
  max_size = Inf,
  drop_unknown = TRUE
)
```

Parameter	Description
`file`	Path to the exported .csv.
`id_col`, `name_col`	Column names holding the pathway object ID and the pathway name.
`gene_col`	Column holding the gene identifier to use. For human analyses with MAGMA this should be an NCBI/Entrez gene ID column, since MAGMA's gene location files are Entrez-keyed.
`min_size`, `max_size`	Pathway size bounds, applied after de-duplication.
`drop_unknown`	Logical. Drop blank and "unknown" gene identifiers.

`catfish_write_set_annot()`

Write a CATFISH pathway data.frame as a MAGMA –set-annot file

```
catfish_write_set_annot(pathways, file)
```

Parameter	Description
`pathways`	data.frame from `catfish_load_gmt` or `catfish_load_biocyc`.
`file`	Output path. One row per pathway: set name followed by its gene identifiers, tab separated.

`catfish_acat_pathways()`

ACAT pathway test on gene-level p-values

```
catfish_acat_pathways(
  gene_results,
  pathways = NULL,
  species = NULL,
  pmn_gene_col = NULL,
  gene_col = "GENE",
```

```
  p_col = "P",
  min_p = 1e-15,
  do_fix = TRUE,
  output = FALSE,
  out_dir = "catfish_acat"
)
```

Parameter	Description
`gene_results`	data.frame with at least a gene ID column and a p-value column. For MAGMA `.genes.out`, these are commonly `GENE` and `P`.
`pathways`	Either: • A named list: each element is a character vector of gene IDs in that pathway, OR • A data.frame with columns `pathway_id`, `gene_id` (and optional `pathway_name`).
`species`	Character. One of "maize", "sorghum", "arabidopsis", "plant", or "fly". If provided, built-in pathways are loaded via `catfish_load_pathways`. Provide either `pathways` OR `species`, but not both.
`pmn_gene_col`	Character or NULL. Passed to `catfish_load_pathways(gene_col=...)` when `species` is used. If NULL, the loader uses default gene column for each species.
`gene_col`	Character. Column name in `gene_results` containing gene IDs. Default `"GENE"`.
`p_col`	Character. Column name in `gene_results` containing gene-level p-values. Default `"P"`.
`min_p`	Numeric. Lower cap for very small p-values before ACAT. Default 1e-15.
`do_fix`	Logical. If TRUE (default), clean/cap p-values with `fix_p_for_acat`.
`output`	Logical. If TRUE, write a CSV of results to `out_dir`.
`out_dir`	Character. Directory to write CSV when `output = TRUE`. Default `"catfish_acat"`.

`catfish_fisher_pathways()`

Pathway-level Fisher test on gene-level p-values

```
catfish_fisher_pathways(
  gene_results,
  pathways = NULL,
  species = NULL,
  pmn_gene_col = NULL,
  gene_col = "GENE",
  p_col = "P",
  min_p = 1e-15,
  do_fix = TRUE,
  output = FALSE,
  out_dir = "catfish_fisher"
)
```

Parameter	Description
`gene_results`	data.frame with at least a gene ID column and a p-value column. For MAGMA `.genes.out`, these are commonly `GENE` and `P`.
`pathways`	Either: • A named list: each element is a character vector of gene IDs in that pathway, OR • A data.frame with columns `pathway_id`, `gene_id` (and optional `pathway_name`).
`species`	Optional; one of "maize", "sorghum", "arabidopsis", "plant", "fly". If provided, built-in pathways are loaded via `catfish_load_pathways()`. Provide either `pathways` OR `species`, but not both.
`pmn_gene_col`	Optional; passed to `catfish_load_pathways(gene_col=...)` when `species` is used. If NULL, the PMN loader prefers "Gene-name" then "Gene-id".
`gene_col`	Column name in `gene_results` containing gene IDs (default "GENE").
`p_col`	Column name in `gene_results` containing gene-level p-values (default "P").
`min_p`	Lower cap for very small p-values (default 1e-15).
`do_fix`	If TRUE (default), clean/cap p-values using `fix_p_for_acat()` to ensure values lie strictly in (0,1).
`output`	If TRUE, write a CSV of results to `out_dir`.
`out_dir`	Directory to write CSV when `output = TRUE` (default "catfish_fisher").

`catfish_soft_tfisher_adaptive_pathways()`

Pathway-level adaptive* soft TFisher (omnibus) on gene-level p-values*

```
catfish_soft_tfisher_adaptive_pathways(
  gene_results,
  pathways = NULL,
  species = NULL,
  pmn_gene_col = NULL,
  gene_col = "GENE",
  p_col = "P",
  tau_grid = "auto",
  min_p = 1e-15,
  do_fix = TRUE,
  output = FALSE,
  out_dir = "catfish_tfisher_soft_adaptive"
)
```

Parameter	Description
`gene_results`	data.frame with at least a gene ID column and a p-value column. For MAGMA `.genes.out`, these are commonly `GENE` and `P`.
`pathways`	Either: • A named list: each element is a character vector of gene IDs in that pathway, OR • A data.frame with columns `pathway_id`, `gene_id` (and optional `pathway_name`).

Parameter	Description
species	Optional; one of “maize”, “sorghum”, “arabidopsis”, “plant”, “fly”. If provided, built-in PMN pathways are loaded via `catfish_load_pathways()`. Provide either `pathways` OR `species`, but not both.
pmn_gene_col	Optional; passed to `catfish_load_pathways(gene_col=...)` when `species` is used. If NULL, the PMN loader prefers “Gene-name” then “Gene-id”.
gene_col	Column name in `gene_results` containing gene IDs (default “GENE”).
p_col	Column name in `gene_results` containing gene-level p-values (default “P”).
tau_grid	Adaptive soft-TFisher thresholds: either a numeric vector of `tau` values (internally sorted increasing, as required by `TFisher::stat.soft.omni()`), or `"auto"` (default) to choose the grid from the empirical quantiles of the pooled gene-level p-values. Values must satisfy 0 < tau < 1; the tau = 1 Fisher corner is excluded here because CATFISH already carries Fisher as a separate component test.
min_p	Lower cap for extremely small p-values before TFisher math (default 1e-15).
do_fix	If TRUE (default), clean/cap p-values using `fix_p_for_acat()` to ensure values lie strictly in (0,1).
output	If TRUE, write a CSV of results to `out_dir`.
out_dir	Directory to write CSV when `output = TRUE` (default “catfish_tfisher_soft_adaptive”).

`catfish_minp_pathways()`

Pathway-level minP (Tippett/Wilkinson) on gene-level p-values

```
catfish_minp_pathways(
  gene_results,
  pathways = NULL,
  species = NULL,
  pmn_gene_col = NULL,
  gene_col = "GENE",
  p_col = "P",
  min_p = 1e-15,
  do_fix = TRUE,
  output = FALSE,
  out_dir = "catfish_minp"
)
```

Parameter	Description
gene_results	data.frame with at least a gene ID column and a p-value column. For MAGMA `.genes.out`, these are commonly `GENE` and `P`.
pathways	Either: • A named list: each element is a character vector of gene IDs in that pathway, OR • A data.frame with columns `pathway_id`, `gene_id` (and optional `pathway_name`).

Parameter	Description
species	Optional; one of “maize”, “sorghum”, “arabidopsis”, “plant”, “fly”. If provided, built-in PMN pathways are loaded via catfish_load_pathways(). Provide either pathways OR species, but not both.
pmn_gene_col	Optional; passed to catfish_load_pathways(gene_col=...) when species is used. If NULL, the PMN loader prefers “Gene-name” then “Gene-id”.
gene_col	Column name in gene_results containing gene IDs (default “GENE”).
p_col	Column name in gene_results containing gene-level p-values (default “P”).
min_p	Lower cap for extremely small p-values before transformations (default 1e-15).
do_fix	If TRUE (default), clean/cap p-values using fix_p_for_acat() to ensure values lie strictly in (0,1).
output	If TRUE, write a CSV of results to out_dir.
out_dir	Directory to write CSV when output = TRUE (default “catfish_minp”).

catfish_stoufferZ_pathways()

Pathway-level (weighted) Stouffer Z test on gene-level Z statistics

```
catfish_stoufferZ_pathways(
  gene_results,
  pathways = NULL,
  species = NULL,
  pmn_gene_col = NULL,
  gene_col = "GENE",
  z_col = "ZSTAT",
  weight_col = NULL,
  min_abs_w = 1e-08,
  min_p = 1e-15,
  alternative = c("greater", "two.sided", "less"),
  output = FALSE,
  out_dir = "catfish_stouffer_z"
)
```

Parameter	Description
gene_results	data.frame with at least a gene ID column and a Z-statistic column. For MAGMA .genes.out, you may have GENE and ZSTAT (or similar).
pathways	Either: • A named list: each element is a character vector of gene IDs in that pathway, OR • A data.frame with columns pathway_id, gene_id (and optional pathway_name).
species	Optional; one of “maize”, “sorghum”, “arabidopsis”, “plant”, “fly”. If provided, built-in PMN pathways are loaded via catfish_load_pathways(). Provide either pathways OR species, but not both.

Parameter	Description
`pmn_gene_col`	Optional; passed to `catfish_load_pathways(gene_col=...)` when `species` is used. If NULL, the PMN loader prefers "Gene-name" then "Gene-id".
`gene_col`	Column name in `gene_results` containing gene IDs (default "GENE").
`z_col`	Column name in `gene_results` containing gene-level Z statistics (default "ZSTAT").
`weight_col`	Optional column name in `gene_results` for per-gene weights (default NULL = equal weights). Examples: `sqrt(NSNPS)`, `1/SE`, etc.
`min_abs_w`	Non-finite / NA / <=0 weights are replaced by this small positive value (default 1e-8).
`min_p`	Lower cap for extremely small p-values to avoid underflow to 0 (default 1e-15).
`alternative`	One of `"greater"`, `"two.sided"`, `"less"`. `"greater"` tests for positive enrichment (default "greater").
`output`	If TRUE, write a CSV of results to `out_dir`.
`out_dir`	Directory to write CSV when `output = TRUE` (default "catfish_stouffer_z").

`catfish_wy_pathways()`

Pathway testing with Westfall–Young family-wise calibration

```
catfish_wy_pathways(
  gene_results,
  pathways = NULL,
  species = NULL,
  pmn_gene_col = NULL,
  gene_col = "GENE",
  p_col = "P",
  chr_col = "CHR",
  z_col = NULL,
  cor_pairs = NULL,
  cor_file = NULL,
  B = 1000L,
  use_tfisher = TRUE,
  tau_grid = c(0.01, 0.05, 0.5),
  min_genes = 10L,
  max_genes = 200L,
  seed = 20260706L,
  verbose = TRUE,
  output = FALSE,
  out_dir = "catfish_wy"
)
```

Parameter	Description
`gene_results`	data.frame of gene-level results (e.g. MAGMA `.genes.out`), with columns for gene ID, p-value and chromosome.
`pathways`	Named list of gene-ID vectors, or a data.frame with `pathway_id` and `gene_id`. Provide this or `species`.

Parameter	Description
species	Optional; built-in PMN pathways via catfish_load_pathways().
pmn_gene_col	Passed to catfish_load_pathways() when species is used.
gene_col, p_col, chr_col	Columns of gene_results holding the gene ID, gene-level p-value and chromosome.
z_col	Optional column (e.g. MAGMA ZSTAT); if given, genes with a non-finite value are dropped, as MAGMA output is usually filtered.
cor_pairs	data.frame of gene pairs and correlations (gene1, gene2, r), e.g. from magma_genesraw_to_cor_pairs_banded().
cor_file	Path to the same table on disk; used if cor_pairs is NULL.
B	Number of genome-wide null replicates.
use_tfisher	Include adaptive soft TFisher as a fifth component.
tau_grid	Fixed soft TFisher truncation grid. The default is the grid of Zhang et al. (2020) without its tau=1 corner, which is Fisher.
min_genes, max_genes	Pathway size limits, counted over genes present in gene_results.
seed	Random seed for the null draws (NULL leaves it unset).
verbose	Print progress.
output	If TRUE, write a CSV to out_dir.
out_dir	Output directory when output = TRUE.

catfish_omni2_pathways()

CATFISH omnibus pathway p-values (analytic, global resampling, or Westfall–Young)

```
catfish_omni2_pathways(
  gene_results,
  pathways = NULL,
  species = NULL,
  pmn_gene_col = NULL,
  gene_col = "GENE",
  p_raw_col = "P",
  p_adj_col = NULL,
  z_col = NULL,
  weight_col = NULL,
  chr_col = NULL,
  tau_grid = "auto",
  tau_cap = 1,
  min_p = 1e-50,
  min_genes = 2L,
  max_genes = Inf,
  use_tfisher = TRUE,
  do_fix = TRUE,
  stouffer_min_abs_w = 1e-08,
  stouffer_alternative = c("greater", "two.sided", "less"),
  magma_out = NULL,
  include_magma_in_omni = TRUE,
  include_magma_in_perm = FALSE,
  omnibus = c("ACAT", "minP"),
  B_perm = NULL,
```

```
  perm_mode = c("none", "global", "wy", "both"),
  wy_tau_grid = c(0.01, 0.05, 0.5),
  tail_mode = c("empirical", "hybrid_gpd"),
  tail_switch_exceed = 10L,
  tail_gpd_k = 250L,
  tail_min_B = 10000L,
  tail_min_tail = 50L,
  B_global = 0L,
  B_wy = 0L,
  perm_pool = c("raw", "obs"),
  magma_cor_file = NULL,
  magma_cor_pairs = NULL,
  seed = NULL,
  output = FALSE,
  out_dir = "catfish_omni2"
)
```

Parameter	Description
`gene_results`	data.frame of gene-level results (one row per gene recommended). Must contain `gene_col` and at least one p-value column (`p_raw_col` and/or `p_adj_col`). If you want Stouffer, it must contain `z_col`. If you want weighted Stouffer, it must contain `weight_col`. For Westfall–Young it must also contain `chr_col`.
`pathways`	Pathway definitions (provide `pathways` OR `species`). Either a data.frame with columns `pathway_id`, `gene_id` (optional `pathway_name`), or a named list mapping `pathway_id -> character vector of genes`.
`species`	Optional character (provide `species` OR `pathways`), one of `"maize"`, `"sorghum"`, `"arabidopsis"`, `"plant"`, `"fly"`, `"human"` (HumanCyc) or `"human_reactome"` (Reactome, Entrez IDs). When provided, pathways are loaded via `catfish_load_pathways()`.
`pmn_gene_col`	Optional character. When `species` is used, passed to `catfish_load_pathways(species=..., gene_col=pmn_gene_col)`.
`gene_col`	Character. Gene ID column in `gene_results`. Default `"GENE"`.
`p_raw_col`	Character. Raw p-value column in `gene_results`. Default `"P"`.
`p_adj_col`	Character or NULL. Preferred/adjusted p-value column. If supplied and present, used for ACAT/Fisher/TFisher/minP. Default NULL.
`z_col`	Character or NULL. Gene Z column for Stouffer. If NULL, Stouffer is skipped. Default NULL.
`weight_col`	Character or NULL. Optional weights for Stouffer. Default NULL (equal weights).
`chr_col`	Character or NULL. Chromosome column in `gene_results`. Required for `perm_mode = "wy"` / `"both"`; genes are drawn jointly within a chromosome and independently across chromosomes. Default NULL.
`tau_grid`	Adaptive soft-TFisher thresholds for the analytic component. Either a numeric vector of tau values, or `"auto"` (default) to select the grid from the empirical quantiles of the pooled gene-level p-values.

Parameter	Description
tau_cap	Numeric. Upper bound for tau_grid (sanity bound). Default 1.
min_p	Numeric. P-values are clipped to [min_p, 1-min_p] to avoid numeric issues. Default 1e-50.
min_genes	Integer. Minimum genes required per pathway. Default 2.
max_genes	Numeric. Maximum genes per pathway (counted over genes present). Default Inf (no upper limit).
use_tfisher	Logical. Include the adaptive soft-TFisher component. Default TRUE.
do_fix	Logical. If TRUE, uses your fix_p_for_acat() if available (else clips). Default TRUE.
stouffer_min_abs_w	Numeric. Non-finite / <=0 Stouffer weights replaced by this. Default 1e-8.
stouffer_alternative	One of "greater", "two.sided", "less" for Stouffer. Default "greater".
magma_out	Optional data.frame of MAGMA competitive results with columns pathway_id, magma_pvalue.
include_magma_in_omni	Logical. Include MAGMA p-values as a component in the omnibus (if provided). Default TRUE.
include_magma_in_perm	Logical. Include MAGMA p-values inside the resampling null. Generally not recommended. If FALSE and permutations are run, MAGMA is omitted from the permuted omnibus. Default FALSE.
omnibus	Character. "ACAT" or "minP" for combining method p-values. Default "ACAT". The Westfall–Young omnibus always uses ACAT.
B_perm	Integer or NULL. Convenience parameter for number of resampling iterations. If provided, it overrides B_global/B_wy according to perm_mode.
perm_mode	Character. One of "none" (analytic only), "global" (global gene resampling), "wy" (Westfall–Young) or "both" (global and Westfall–Young; omni_p_final prefers the Westfall–Young calibrated p-value).
wy_tau_grid	Numeric. Fixed soft-TFisher grid used inside the Westfall–Young omnibus. Default c(0.01, 0.05, 0.5): the grid of Zhang et al. (2020) without its tau = 1 corner, which is Fisher.
tail_mode	Character. "empirical" or "hybrid_gpd" (GPD tail extrapolation for global-resampling p-values below the Monte Carlo floor). Default "empirical".
tail_switch_exceed	Integer. Switch to GPD extrapolation when the number of null exceedances falls below this. Default 10.
tail_gpd_k	Integer. Number of extreme null draws used to fit the GPD. Default 250.
tail_min_B	Integer. Minimum number of null draws required before GPD is attempted. Default 10000.
tail_min_tail	Integer. Minimum tail observations required for a stable GPD fit. Default 50.
B_global	Integer. Number of global-resampling iterations (used when B_perm is NULL).
B_wy	Integer. Number of genome-wide Westfall–Young null replicates (used when B_perm is NULL).
perm_pool	Character. For global resampling only: "raw" (use raw p-values) or "obs" (use p used by methods).

Parameter	Description
magma_cor_file	Character or NULL. Path to 3-col correlation pairs file: gene1 gene2 r. Used by Westfall–Young; without it (or magma_cor_pairs) genes are treated as independent.
magma_cor_pairs	data.frame or NULL. In-memory correlation pairs with columns gene1, gene2, r. If provided, it overrides magma_cor_file.
seed	Integer or NULL. Random seed.
output	Logical. If TRUE, writes CSV to out_dir and attaches "file" attribute. Default FALSE.
out_dir	Character. Output directory. Default "catfish_omni2".

gene_prioritize()

Prioritise candidate genes from GWAS, MAGMA and pathway evidence

```
gene_prioritize(
  magma,
  pathway_results,
  pathways = NULL,
  species = NULL,
  pmn_gene_col = NULL,
  gwas = NULL,
  gene_annot = NULL,
  gene_loc = NULL,
  magma_gene_col = "GENE",
  magma_p_col = NULL,
  pathway_id_col = "pathway_id",
  pathway_stat_col = NULL,
  gwas_snp_col = "SNP",
  gwas_p_col = NULL,
  gwas_neglog10p_col = NULL,
  gwas_chr_col = "CHR",
  gwas_pos_col = "BP",
  symbol_col = NULL,
  window = 10000L,
  w_gwas = 1,
  w_magma = 1,
  w_pathway = 0.5,
  w_hub = 0.3,
  targets = NULL,
  chunk = 2000L,
  verbose = TRUE,
  output = FALSE,
  out_file = "gene_prioritize.csv"
)
```

Parameter	Description
magma	MAGMA gene results: data.frame or file path.
pathway_results	Pathway results: data.frame or file path (CSV/TSV).
pathways	Pathway membership: list, data.frame or set-file path. Give this or species.

Parameter	Description
species	Built-in pathway set (see catfish_load_pathways); give this or pathways.
pmn_gene_col	Gene-ID column of the built-in set, passed to catfish_load_pathways(gene_col = ...) when species is used.
gwas	Optional SNP-level results: data.frame or file path.
gene_annot	Optional SNP-to-gene annotation: MAGMA .genes.annot path or data.frame with gene and SNP.
gene_loc	Optional gene locations: .gene.loc path or data.frame.
magma_gene_col, magma_p_col	Gene ID and p-value columns of magma.
pathway_id_col, pathway_stat_col	Pathway ID and statistic columns of pathway_results.
gwas_snp_col, gwas_p_col, gwas_neglog10p_col, gwas_chr_col, gwas_pos_col	Columns of gwas.
symbol_col	Column of gene_loc holding gene symbols: a name, or a number for a header-less .gene.loc file (MAGMA's NCBI files keep the symbol in column 6). NULL uses a symbol column if present, otherwise column 6 of a 6-column file.
window	Base pairs added on each side of a gene for the position route. Default 10000.
w_gwas, w_magma, w_pathway, w_hub	Layer weights. Defaults 1, 1, 0.5, 0.3.
targets	Optional character vector of known genes (IDs or symbols) to report as positive controls.
chunk	Genes read at a time from a .genes.annot file.
verbose	Print progress.
output	If TRUE, write the table to out_file.
out_file	CSV path used when output = TRUE.

get_gene_lengths()

Extract gene lengths from a GFF3 file

```
get_gene_lengths(
  gff3_file,
  feature = "gene",
  id_key = "ID",
  keep_chr_prefix = NULL,
  output = FALSE,
  output_dir = ".",
  file_name = "gene_lengths.tsv"
)
```

Parameter	Description
gff3_file	Path to a GFF3 file (.gff3 or .gff3.gz).
feature	Which feature type to treat as "gene" (default "gene").
id_key	Attribute key used for gene ID (default "ID").
keep_chr_prefix	If non-NULL, enforce a prefix in CHR (e.g. "chr"). If NULL, leaves seqid as-is.
output	Logical; if TRUE write TSV to output_dir/file_name.
output_dir	Directory for output if output=TRUE.
file_name	Output TSV file name if output=TRUE.

catfish_adjust_gene_p()

Adjust MAGMA gene p-values using gene length and NSNPS

```
catfish_adjust_gene_p(
  gene_results,
  gene_lengths,
  gene_col = "GENE",
  nsnp_col = "NSNPS",
  p_col = "P",
  z_col = "ZSTAT",
  len_gene_col = "gene_id",
  len_col = "length",
  log1p_covars = TRUE
)
```

Parameter	Description
gene_results	data.frame. Gene-level results from MAGMA (typically from reading a `.genes.out` file). Must contain gene ID, p-value, and NSNPS columns.
gene_lengths	data.frame. Gene lengths, typically from `get_gene_lengths`. Must contain gene ID and length columns.
gene_col	character. Column name in `gene_results` for gene IDs. Default "GENE".
nsnp_col	character. Column name in `gene_results` for the number of SNPs per gene. Default "NSNPS".
p_col	character. Column name in `gene_results` for raw p-values. Default "P".
z_col	character or NULL. Column name for Z-scores in `gene_results`. If NULL or not present, Z-scores are reconstructed from p-values (magnitude only). Default "ZSTAT".
len_gene_col	character. Column name in `gene_lengths` for gene IDs. Default "gene_id".
len_col	character. Column name in `gene_lengths` for gene length. Default "length".
log1p_covars	logical. If TRUE (default), apply log(1+x) transformation to NSNPS and gene length covariates before regression.

`fix_p_for_acat()`

Clean p-values for ACAT

```
fix_p_for_acat(p, min_p = 1e-15)
```

Parameter	Description
p	Numeric vector of p-values (NA allowed). Names are preserved.
min_p	Numeric. Lower cap for very small p-values (default 1e-15).